NB: Open with an advanced pdf reader (e.g., Acrobat to have animations)

Lecture on Conformal Prediction (1/4)

Margaux Zaffran

October 29-30, 2026

LIDAM – ISBA seminar series

FONDATION MATHÉMATIQUE JACQUES HADAMARD

Laboratoire de Mathématiques d'Orsay

LIDAM | Louvain Institute of Data Analysis and Modeling in economics and statistics

ISBA

Why a lecture on conformal prediction?

- Conformal Prediction comes from the 90s, and has emerged as a popular topic in the early 2020's.
  - 1996-1999: emergence of Conformal Prediction (CP) in the historical machine learning community, by Vladimir Vovk, Alexander Gammerman, Vladimir Vapnik, Glenn Shafer
  - 2012+: Jing Lei, Larry Wasserman and colleagues at Carnegie Mellon University (CMU) bring it to the statistics community
  - 2020+: explosion, notably thanks to Rina Barber, Emmanuel Candès, Aaditya Ramdas and Ryan J. Tibshirani

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  Successfully applied to
  - Medical applications
  - Markets / demand forecasting
  - Computer Vision

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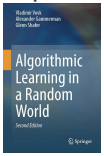
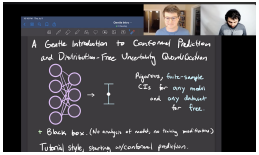
- Because I believe that conformal methods are **important** tools, whose strengths and limitations are sometimes misunderstood.
- To be part of the **diffusion** effort that many colleagues are making.



Book reference: Vovk et al. (2005)
(new edition in 2022)



A gentle tutorial: Angelopoulos and Bates (2023)
+ Videos playlist

R. J. Tibshirani
introductive lecture's notes

Forthcoming book: Angelopoulos, Barber, Bates (2024)
arXiv version

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Aymeric Dieuleveut

Goals and disclaimers

Goals

- Provide a detailed introduction to the basics
- Demystify the results: fair introduction with limits
- Give you insights on how to leverage those techniques in your own fields

Disclaimers

- Many people contributed to the domain - list of references may not be exhaustive
- Multiple other excellent resources

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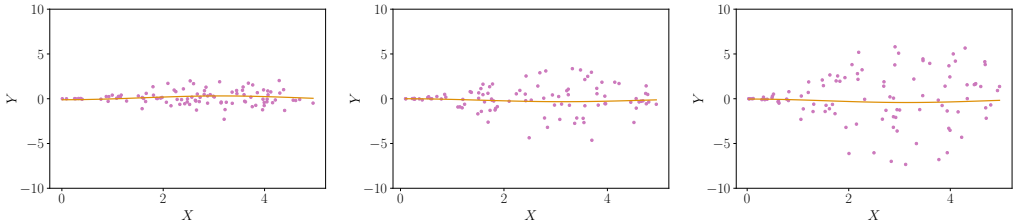

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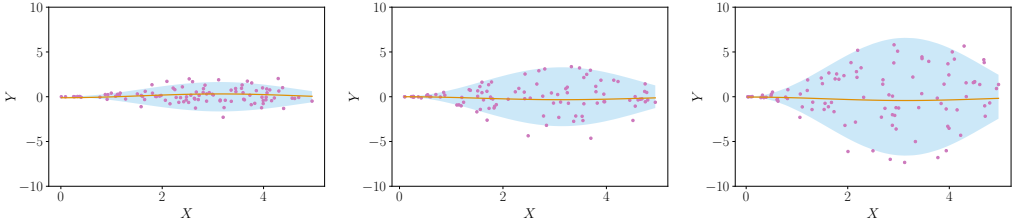


↪ Same "best" predictor, yet 3 distinct underlying phenomena!

$\Longrightarrow$ Quantifying uncertainty conveys this information.

Introductory remarks

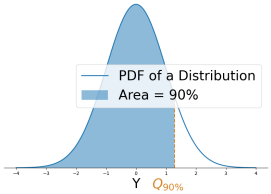
Exchangeability

Split Conformal Prediction (SCP)

On the design choices of conformity scores and (empirical) conditional guarantees

Reminder about quantiles

- Quantile level $\beta \in [0, 1]$
- $Q_Y(\beta) := \inf\{t \in \mathbb{R}, \mathbb{P}(Y \leq t) \geq \beta\}$



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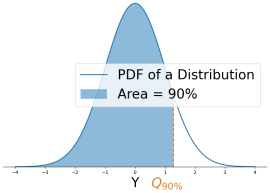

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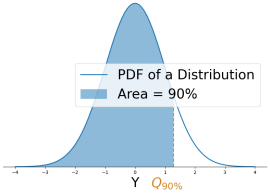

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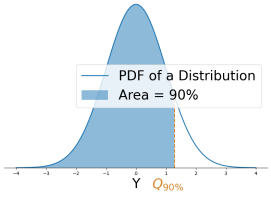

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Example (a special quantile: the median).

$\beta = 0.5$

$\hookrightarrow$ $q_{0.5}(Y_1, \ldots, Y_n)$ is the empirical median of $(Y_1, \ldots, Y_n)$;

$\hookrightarrow$ $Q_Y(0.5)$ represents the median of the distribution of Y.

$\hookrightarrow$ $Q_Y(0.5) = \arg\min_c \mathbb{E}[|Y - c|]$.

Median regression

- The Bayes predictor depends on the chosen loss function.
  $\hookrightarrow$ Bayes predictor $f^\star \in \underset{f}{\operatorname{argmin}} \, \text{Risk}_\ell(f)$

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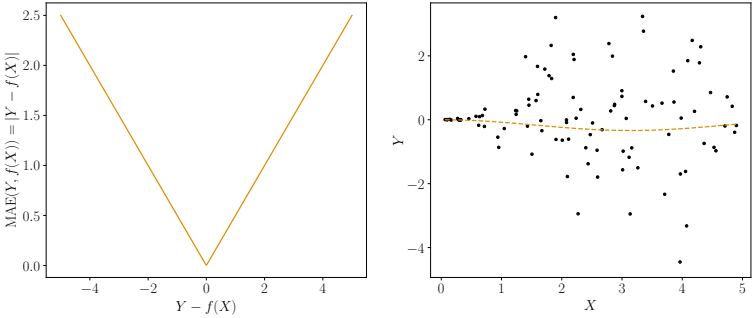

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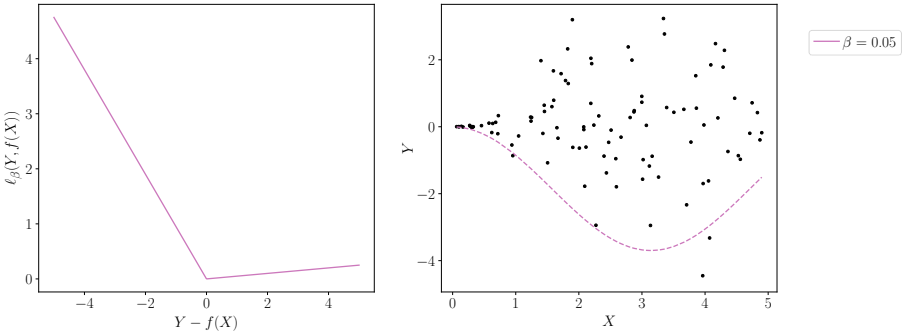

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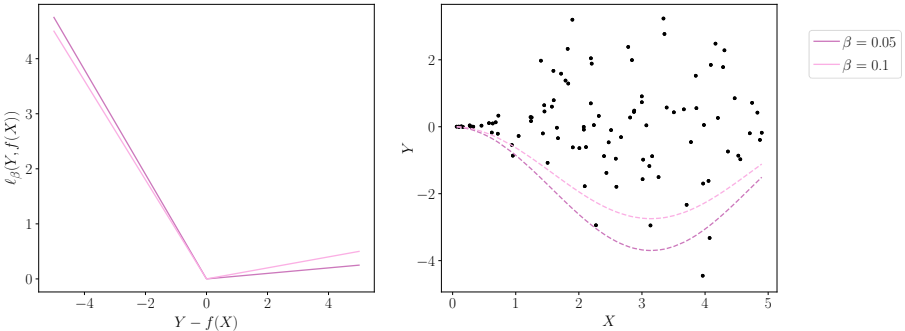

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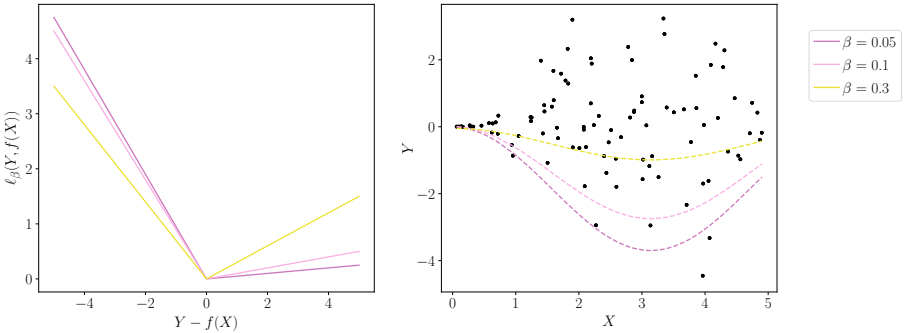

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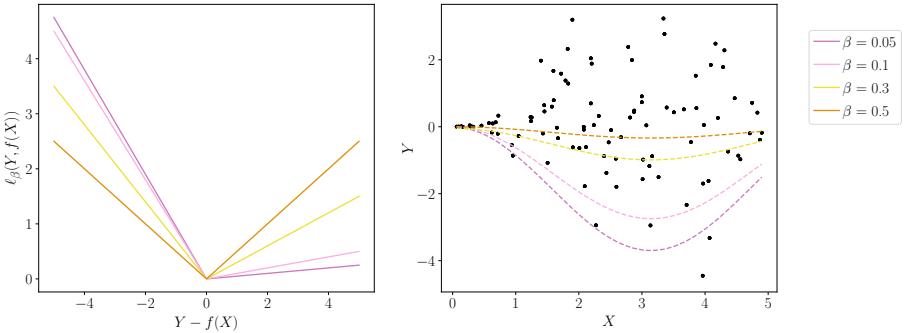

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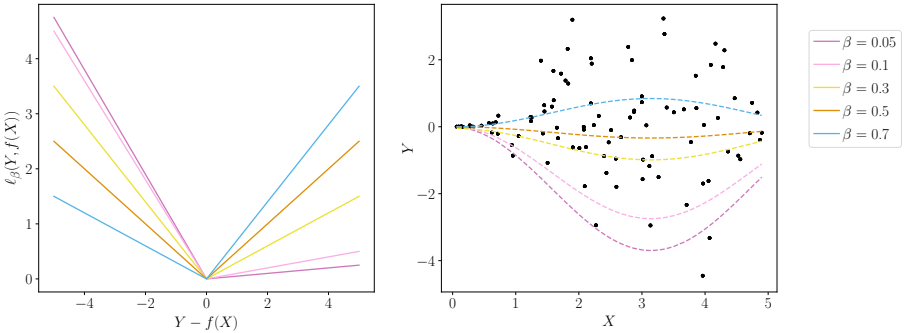

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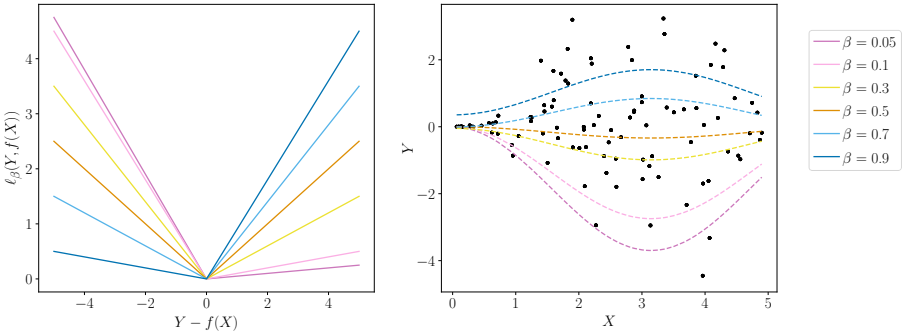

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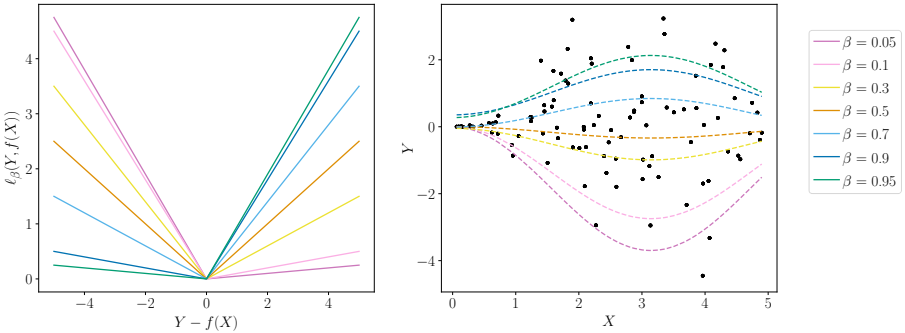

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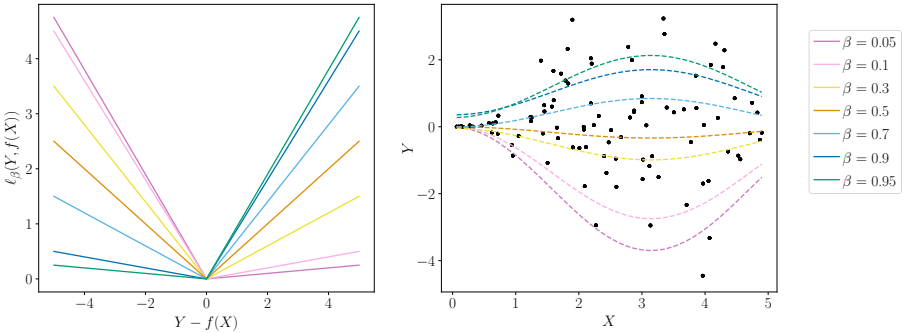

Quantifying predictive uncertainty

- $(X, Y) \in \mathcal{X} \times \mathcal{Y}$ random variables
- n training samples $(X_i, Y_i)_{i=1}^n$
- Goal: predict an unseen point Y_{n+1} at X_{n+1} with **confidence**
- How? Given a miscoverage level $\alpha \in [0, 1]$, build a predictive set $\mathcal{C}_\alpha$ such that:

$$\mathbb{P}\left\{Y_{n+1} \in \mathcal{C}_\alpha\left(X_{n+1}\right)\right\} \geq 1 - \alpha, \tag{1}$$

and $\mathcal{C}_\alpha$ should be as small as possible, in order to be informative

For example: $\alpha = 0.1$ and obtain a 90% coverage interval

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Post-hoc design (black-box view)

We aim for an approach that works in a post-hoc fashion on top of *any* $\mathcal{A}$, with no impact of the choice of $\mathcal{A}$ on the validity of the method.

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  For example: $\alpha = 0.1$ and obtain a 90% coverage interval
- Construction of the predictive intervals should be
  - agnostic to the model
  - agnostic to the data distribution
- *Validity* should be ensured
  - in finite samples
  - for all data distribution and underlying model

Attempt to define distribution-free validity

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

Definition ((lofty) Distribution-free validity)**.**

$\widehat{C}_\alpha$ achieves (lofty) distribution-free validity if for any joint distribution $\mathcal{D}_{n+1}$ on $(\mathcal{X} \times \mathcal{Y})^{n+1}$, we have that:

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✓ What is achievable is to restrict $\mathcal{D}_{n+1}$ to any product distribution $\mathcal{D}^{\otimes(n+1)}$ (i.i.d.) without restriction on $\mathcal{D}$ (distribution on $\mathcal{X} \times \mathcal{Y}$).

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- In fact, we can even weaken this to exchangeable distributions.

This is what we are jumping into!

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2. Split conformal prediction

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1. On exchangeability (theory)
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3. Towards conditional coverage? (theory)
4. Beyond exchangeability
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2. Split conformal prediction (methods) (theory)
3. Towards conditional coverage? (theory)
4. Beyond exchangeability (methods)
5. Computational and statistical trade-offs (methods) (theory)
6. Handling missing data (methods)

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2. Split conformal prediction (methods) (theory)

Introductory remarks

Exchangeability

Split Conformal Prediction (SCP)

On the design choices of conformity scores and (empirical) conditional guarantees

Definition (Exchangeability)**.**

$(Z_i)_{i=1}^n$, a random vector taking values in $\mathcal{Z}^n$, is exchangeable if, for any permutation σ of $[\![1, n]\!]$:

$$(Z_1, \ldots, Z_n) \overset{d}{=} \left(Z_{\sigma(1)}, \ldots, Z_{\sigma(n)}\right).$$

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In particular, exchangeability implies that all the entries of $(Z_i)_{i=1}^n$ are identically distributed.

Some examples of exchangeable random vectors

Example (exchangeable sequences)**.**

- $Z_1, \ldots, Z_n \overset{i.i.d.}{\sim} \mathcal{D}$

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- $Z_1, \ldots, Z_n \overset{i.i.d.}{\sim} \mathcal{D}$
- The components of $\mathcal{N}\left(\begin{pmatrix} m \\ \vdots \\ \vdots \\ m \end{pmatrix}, \begin{pmatrix} \sigma^2 & & & \\ & \ddots & \gamma^2 & \\ & \gamma^2 & \ddots & \\ & & & \sigma^2 \end{pmatrix}\right)$

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- Sampling without replacement from $\{z_1, \ldots, z_n\}$

Characterization in a countable space

Denote p the probability mass function of the joint distribution of $(Z_1, \ldots, Z_n)$.
$(Z_1, \ldots, Z_n)$ are exchangeable if and only if, for any permutation σ of $[\![1, n]\!]$, for any $(z_1, \ldots, z_n) \in \mathcal{Z}^n$ we have:

$$p(z_1, \ldots, z_n) = p\left(z_{\sigma(1)}, \ldots, z_{\sigma(n)}\right).$$

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Characterization in a continuous space

Assume $(Z_1, \ldots, Z_n)$ admits a probability density function, denoted f.
$(Z_1, \ldots, Z_n)$ are exchangeable if and only if, for any permutation σ of $[\![1, n]\!]$, for almost any $(z_1, \ldots, z_n) \in \mathcal{Z}^n$ we have:

$$f(z_1, \ldots, z_n) = f\left(z_{\sigma(1)}, \ldots, z_{\sigma(n)}\right).$$

Interesting properties of exchangeability

Blackboard time!

Achievable definition of distribution-free validity

$\widehat{C}_\alpha$ = estimated predictive set based on n data points.

Definition (Distribution-free validity)**.**

$\widehat{C}_\alpha$ achieves distribution-free validity if:

- for any distribution $\mathcal{D}$,
- for any associated exchangeable joint distribution $\mathcal{D}^{\text{exch}(n+1)}$,

we have that:

$$\mathbb{P}_{\mathcal{D}^{\text{exch}(n+1)}}\left(Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right)\right) \geq 1-\alpha.$$

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- Standard regression case
- Conformalized Quantile Regression (CQR)
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Split Conformal Prediction (SCP) (Vovk et al., 2005)

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SCP: theoretical guarantees

SCP enjoys finite sample guarantees proved in Vovk et al. (2005); Lei et al. (2018).

Theorem (Marginal validity)**.**

Suppose $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable[a]. SCP applied on $(X_i, Y_i)_{i=1}^{n}$ outputs $\widehat{C}_\alpha(\cdot)$ such that:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})\right\} \geq 1 - \alpha.$$

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Additionally, if the scores $\{S_i\}_{i \in \mathrm{Cal}} \cup \{S_{n+1}\}$ are a.s. distinct:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})\right\} \leq 1 - \alpha + \frac{1}{\#\mathrm{Cal} + 1}.$$

[a] Only the calibration and test data need to be exchangeable.

Proof architecture of SCP guarantees

Blackboard time!

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✓ Distribution free, model (regressor) free, finite sample average validity guarantee.

Standard mean-regression SCP – strength: validity – good vs bad estimator

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Additionally, if the scores $\{S_i\}_{i \in \mathrm{Cal}} \cup \{S_{n+1}\}$ are almost surely distinct:

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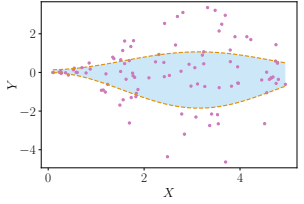
[a] Only the calibration and test data need to be exchangeable.

✗ Marginal coverage: $\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) \,|\, \cancel{X_{n+1} = x}\right\} \geq 1 - \alpha$

Conditional coverage implies adaptiveness

no coverage



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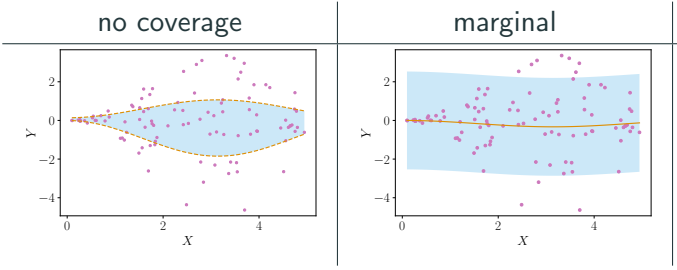
- Marginal coverage: $\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha\left(X_{n+1}\right)\right\}$ the errors may differ across regions of the input space (i.e. non-adaptive)



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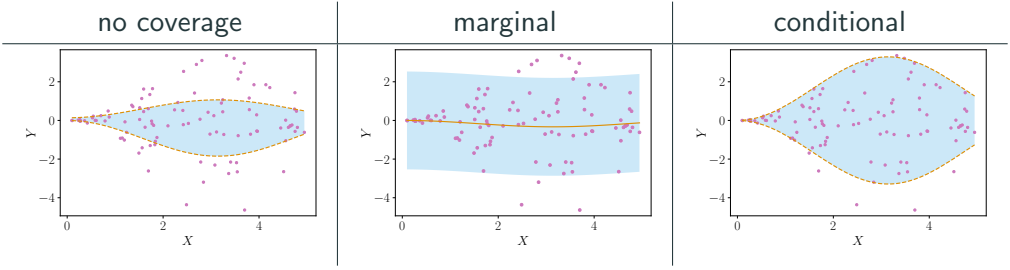

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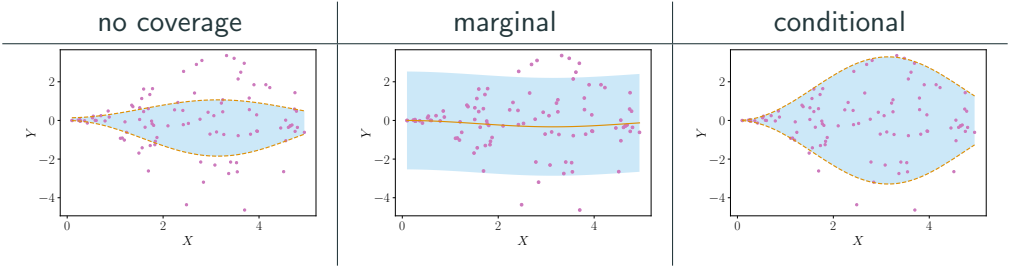
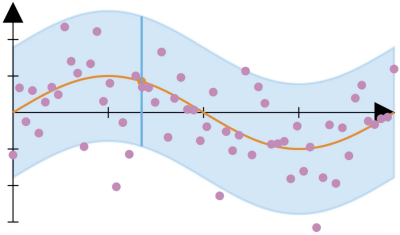
- Conditional coverage is stronger than marginal coverage



Standard mean-regression SCP – weakness: not adaptive



- Predict with $\hat{\mu}$
- Build $\widehat{C}_\alpha(x)$: $[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$

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Conformalized Quantile Regression (CQR) (Romano et al., 2019)

CQR: under vs over coverage

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This procedure enjoys the finite sample guarantee proposed and proved in Romano et al. (2019).

Theorem (Marginal validity of CQR Romano et al. (2019)).

Suppose $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable[a]. CQR on $(X_i, Y_i)_{i=1}^{n}$ outputs $\widehat{C}_\alpha(\cdot)$ such that:

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$$\mathcal{S} = \{S_i = \mathsf{s}\left(X_i, Y_i; \hat{A}\right), i \in \mathrm{Cal}\} \cup \{+\infty\}$$

Ex 1: $\mathsf{s}(\hat{A}(X_i), Y_i) := |\hat{\mu}(X_i) - Y_i|$ in regression with standard scores

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4. Compute the $1-\alpha$ quantile of these scores, noted $q_{1-\alpha}(\mathcal{S})$

SCP is defined by the conformity score function

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5. For a new point X_{n+1}, return

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$\hookrightarrow$ The definition of the conformity scores is crucial, as they incorporate almost all the information: data + underlying model

SCP: theoretical guarantees

This procedure enjoys the finite sample guarantee proposed and proved in Vovk et al. (2005).

Theorem (Marginal validity of SCP Vovk et al. (2005)).

Suppose $(X_i, Y_i)_{i=1}^{n+1}$ are exchangeable[a]. SCP on $(X_i, Y_i)_{i=1}^{n}$ outputs $\widehat{C}_\alpha(\cdot)$ such that:

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})\right\} \geq 1 - \alpha.$$

If, in addition, the scores $\{S_i\}_{i \in \mathrm{Cal}} \cup \{S_{n+1}\}$ are almost surely distinct, then

$$\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1})\right\} \leq 1 - \alpha + \frac{1}{\#\mathrm{Cal} + 1}.$$

[a] Only the calibration and test data need to be exchangeable.

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✗ Marginal coverage: $\mathbb{P}\left\{Y_{n+1} \in \widehat{C}_\alpha(X_{n+1}) \,|\, X_{n+1} = x\right\} \geq 1 - \alpha$

Introductory remarks

Exchangeability

Split Conformal Prediction (SCP)

- Standard regression case
- Conformalized Quantile Regression (CQR)
- Generalization of SCP: going beyond regression
- Some examples of classification scores

On the design choices of conformity scores and (empirical) conditional guarantees

SCP: standard classification

- $Y \in \{1, \ldots, C\}$ (C classes)
- $\hat{A}(X) = (\hat{p}_1(X), \ldots, \hat{p}_C(X))$ (estimated probabilities)

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SCP: standard classification in practice

Ex: $Y_i \in \{\text{“dog”}, \text{“tiger”}, \text{“cat”}\}$, with $\alpha = 0.1$

- Scores on the calibration set

Cal_i	“dog”	“dog”	“dog”	“tiger”	“tiger”	“tiger”	“tiger”	“cat”	“cat”	“cat”
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- $\widehat{C}_\alpha(X_{n+1}) = \{\text{“tiger”}, \text{“cat”}\}$

SCP: standard classification in practice, cont'd

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$\hat{p}_{\text{cat}}(X_i)$	0.03	0.05	0.05	0.10	0.15	0.20	0.25	0.65	0.60	0.55
S_i	0.05	0.1	0.15	0.15	0.20	0.25	0.30	0.35	0.40	0.45

- $q_{1-\alpha}(\mathcal{S}) = 0.45$
- $\hat{A}(X_{n+1}) = (0.05, 0.60, 0.35)$
  - $\hookrightarrow$ s$(\hat{A}(X_{n+1}), \text{“dog”}) = 0.95$ $\quad$ “dog” $\notin \widehat{C}_\alpha(X_{n+1})$
  - $\hookrightarrow$ s$(\hat{A}(X_{n+1}), \text{“tiger”}) = 0.40 \leq q_{1-\alpha}(\mathcal{S})$ $\quad$ “tiger” $\in \widehat{C}_\alpha(X_{n+1})$
  - $\hookrightarrow$ s$(\hat{A}(X_{n+1}), \text{“cat”}) = 0.65$ $\quad$ “cat” $\notin \widehat{C}_\alpha(X_{n+1})$
- $\widehat{C}_\alpha(X_{n+1}) = \{\text{“tiger”}\}$

SCP: limits of the standard classification case

Efficiency yet non-adaptivity of the simplest classification scores

- ✓ Outputs the most efficient set possible (i.e. achieving the smallest average set size, Sadinle et al., 2018),
- ✗ Does not allow to discriminate between "easy" and "hard" test point. In practice, it leads to predictive sets that under-cover (resp. over-cover) on "hard" (resp. "easy") subgroups.
  This is due to the fact that only the prediction for the true class is used to construct the scores.

SCP: classification with Adaptive Prediction Sets[8]

1. Sort in decreasing order $\hat{p}_{\sigma_x(1)}(x) \geq \ldots \geq \hat{p}_{\sigma_x(C)}(x)$

[8] Romano et al. (2020), *Classification with Valid and Adaptive Coverage*, NeurIPS

SCP: classification with Adaptive Prediction Sets[8]

1. Sort in decreasing order $\hat{p}_{\sigma_x(1)}(x) \geq \ldots \geq \hat{p}_{\sigma_x(C)}(x)$
2. $\mathsf{s}(x, y; \hat{p}) := \sum_{k=1}^{\sigma_x^{-1}(y)-1} \hat{p}_{\sigma_x(k)}(x)$ (sum of the estimated probabilities associated to classes at least as large as that of the true class Y)

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3. Return the set of classes $\left\{\sigma_{X_{n+1}}(1), \ldots, \sigma_{X_{n+1}}(r^\star)\right\}$, where

$$r^\star = \underset{1 \leq r \leq C}{\arg\max} \left\{ \sum_{k=1}^{r} \hat{p}_{\sigma_{X_{n+1}}(k)}(X_{n+1}) < q_{1-\alpha}(\mathcal{S}) \right\} + 1$$

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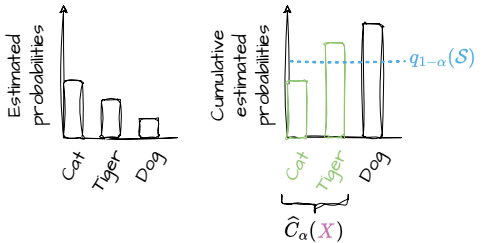

[8] Romano et al. (2020), *Classification with Valid and Adaptive Coverage*, NeurIPS
Figure highly inspired by Angelopoulos and Bates (2023).

SCP: classification with Adaptive Prediction Sets in practice

Ex: $Y \in \{\text{“dog”}, \text{“tiger”}, \text{“cat”}\}$, with $\alpha = 0.1$

- Scores on the calibration set

Cal_i	“dog”	“dog”	“dog”	“tiger”	“tiger”	“tiger”	“tiger”	“cat”	“cat”	“cat”
$\hat{\rho}_{\text{dog}}(X_i)$	0.95	0.90	0.85	0.05	0.05	0.05	0.10	0.05	0.10	0.15
$\hat{\rho}_{\text{tiger}}(X_i)$	0.02	0.05	0.10	0.85	0.80	0.75	0.75	0.50	0.30	0.30
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S_i	0	0	0	0	0	0	0	0.5	0	0

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- $q_{1-\alpha}(\mathcal{S}) = 0.5$

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$$\widehat{C}_\alpha(X_{n+1}) = \{\text{“tiger”}, \text{“cat”}\}$$

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  $\hookrightarrow$ Ex 2: $\hat{A}(X_{n+1}) = (0.03, 0.95, 0.02)$

SCP: classification with Adaptive Prediction Sets in practice

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  $$\widehat{C}_\alpha(X_{n+1}) = \{\text{“tiger”}, \text{“cat”}\}$$

  $\hookrightarrow$ Ex 2: $\hat{A}(X_{n+1}) = (0.03, 0.95, 0.02), r^\star = 1$

  $$\widehat{C}_\alpha(X_{n+1}) = \{\text{“tiger”}\}$$

Split Conformal Prediction: summary

- **Simple** procedure which quantifies the uncertainty of **any** predictive model $\hat{A}$ by returning predictive regions
- **Finite-sample** guarantees
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  $\hookrightarrow$ marginal also over the whole calibration set and the test point!

Side comment: it is possible to achieve exactly $1-\alpha$ coverage!

Note this other view on conformal sets, through conformal p-values:

$$\begin{aligned}
\widehat{C}_\alpha(x) &= \left\{ y \in \mathcal{Y}, s\left(x, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S}) \right\} \\
&= \left\{ y \in \mathcal{Y}, \sum_{i \in \mathrm{Cal}} \mathbb{1}\left\{ s\left(x, y; \hat{A}\right) > S_i \right\} < (1-\alpha)(1+\#\mathrm{Cal}) \right\} \\
&= \left\{ y \in \mathcal{Y}, \frac{1 + \sum_{i \in \mathrm{Cal}} \mathbb{1}\left\{ s\left(x, y; \hat{A}\right) \leq S_i \right\}}{1+\#\mathrm{Cal}} > \alpha \right\}.
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The randomization procedures drawing $U \sim \mathcal{U}([0,1])$ and outputting: $\widehat{C}^{\mathrm{r}}_\alpha(x) :=$

$$\left\{y \in \mathcal{Y}, \frac{\sum_{i \in \mathrm{Cal}} \mathbb{1}\left\{s\left(x, y; \hat{A}\right) < S_i\right\} + U\left(1 + \sum_{i \in \mathrm{Cal}} \mathbb{1}\left\{s\left(x, y; \hat{A}\right) = S_i\right\}\right)}{\#\mathrm{Cal}+1} > \alpha\right\},$$

achieves exactly $1-\alpha$ coverage under exchangeability (smoothed conformal prediction, Vovk et al., 2005).

Introductory remarks

Exchangeability

Split Conformal Prediction (SCP)

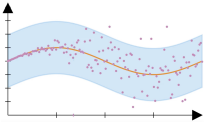
On the design choices of conformity scores and (empirical) conditional guarantees

SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathsf{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}\left(\mathcal{S}\right)\}$$

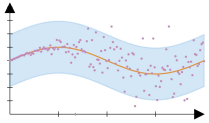
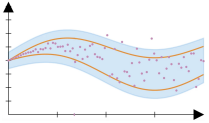
SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)		
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$		
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$		
Visu.			
✓	black-box around a "usable" prediction		
✗	not adaptive		

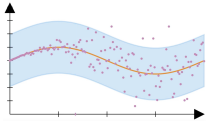
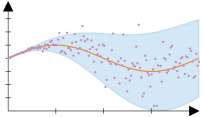
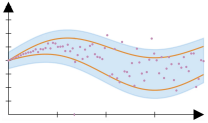
SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)		**CQR** Romano et al. (2019)
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$		$\max(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X) - Y,$ $Y - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X))$
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$		$[\widehat{\mathrm{QR}}_{\mathrm{lower}}(x) - q_{1-\alpha}(\mathcal{S});$ $\widehat{\mathrm{QR}}_{\mathrm{upper}}(x) + q_{1-\alpha}(\mathcal{S})]$
Visu.			
✓	black-box around a "usable" prediction		adaptive
✗	not adaptive		no black-box around a "usable" prediction

SCP: what choices for the regression scores?

$$\widehat{C}_\alpha(X_{n+1}) = \{y \text{ such that } \mathrm{s}\left(X_{n+1}, y; \hat{A}\right) \leq q_{1-\alpha}(\mathcal{S})\}$$

	Standard SCP Vovk et al. (2005)	**Locally weighted SCP** Lei et al. (2018)	**CQR** Romano et al. (2019)
$\mathrm{s}(\hat{A}(X), Y)$	$\lvert\hat{\mu}(X) - Y\rvert$	$\frac{\lvert\hat{\mu}(X) - Y\rvert}{\hat{\rho}(X)}$	$\max(\widehat{\mathrm{QR}}_{\mathrm{lower}}(X) - Y,$ $Y - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X))$
$\widehat{C}_\alpha(x)$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$	$[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})\hat{\rho}(x)]$	$[\widehat{\mathrm{QR}}_{\mathrm{lower}}(x) - q_{1-\alpha}(\mathcal{S});$ $\widehat{\mathrm{QR}}_{\mathrm{upper}}(x) + q_{1-\alpha}(\mathcal{S})]$
Visu.			
✓	black-box around a "usable" prediction	black-box around a "usable" prediction	adaptive
✗	not adaptive	limited adaptiveness	no black-box around a "usable" prediction

Challenges that we are going to explore in the next lectures

1. On exchangeability (theory)
2. Split conformal prediction (methods) (theory)

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5. Computational and statistical trade-offs (methods) (theory)
6. Handling missing data (methods)

Some (other, non-exhaustives!) current open directions

- On conformal prediction efficiency (Bars and Humbert, 2025)
- Selective inference (Marandon et al., 2024a)
- Transductive conformal prediction, outlier detection (Vovk et al., 2003; Bates et al., 2023; Marandon et al., 2024b; Gazin et al., 2024)
- Aggregating predictive sets (Gasparin and Ramdas, 2024b,a; Gasparin et al., 2024)
- Multivariate outputs (Klein et al., 2025; Thurin et al., 2025; Kondratyev et al., 2025; Ndiaye, 2025)
- Beyond the indicator loss (Angelopoulos et al., 2022a; Bates et al., 2021; Angelopoulos et al., 2023; Lekeufack et al., 2024)
- Decision-theoretic foundations (Kiyani et al., 2025)

Some (other, non-exhaustives!) current open directions, cont'd

- CP is a very active field of research. Many developments focus on **adapting CP to specific frameworks**, such as:
  - Survival Analysis (Candès et al., 2023),
  - Causal Inference (Lei and Candès, 2021; Jin et al., 2023),
  - NLP (Schuster et al., 2022),
  - RL (Taufiq et al., 2022),
  - federated learning (Humbert et al., 2023; Plassier et al., 2023; Lu et al., 2023),
  - applications (medical (Angelopoulos et al., 2022b; Lu et al., 2022), energy (Kath and Ziel, 2021), etc.) and more.

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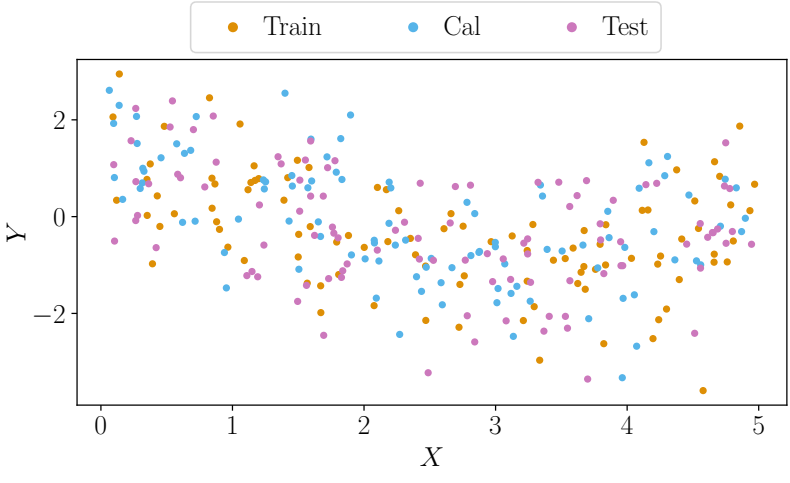
SCP

CQR

Split Conformal Prediction (SCP)[1,2,3]: toy example

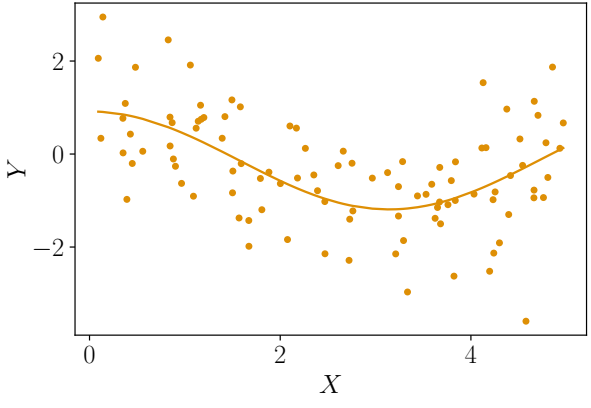


[1] Vovk et al. (2005), *Algorithmic Learning in a Random World*

[2] Papadopoulos et al. (2002), *Inductive Confidence Machines for Regression*, ECML

[3] Lei et al. (2018), *Distribution-Free Predictive Inference for Regression*, JRSS B

Split Conformal Prediction (SCP)[1,2,3]: toy example — training step



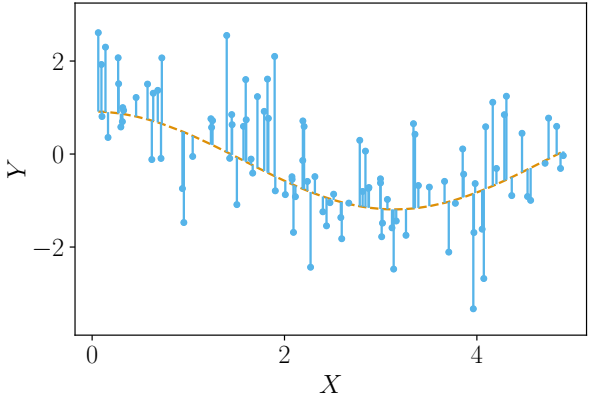
- Learn (or get) $\hat{\mu}$

[1] Vovk et al. (2005), *Algorithmic Learning in a Random World*

[2] Papadopoulos et al. (2002), *Inductive Confidence Machines for Regression*, ECML

[3] Lei et al. (2018), *Distribution-Free Predictive Inference for Regression*, JRSS B

Split Conformal Prediction (SCP)[1,2,3]: toy example — calibration step



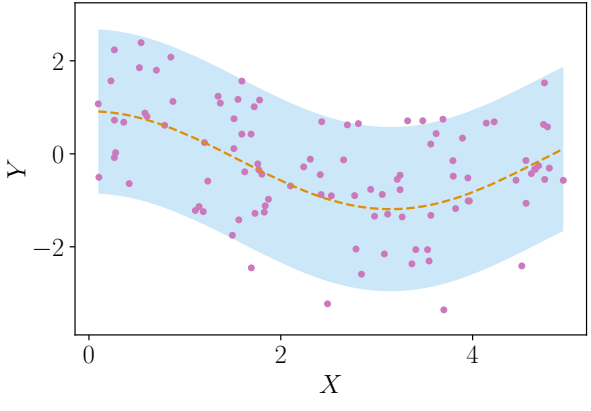
- Predict with $\hat{\mu}$
- Get the $|\text{residuals}|$, a.k.a. conformity scores
- Compute the $(1-\alpha)$ empirical quantile of $\mathcal{S} = \{|\text{residuals}|\}_{\text{Cal}} \cup \{+\infty\}$, noted $q_{1-\alpha}(\mathcal{S})$

[1] Vovk et al. (2005), *Algorithmic Learning in a Random World*

[2] Papadopoulos et al. (2002), *Inductive Confidence Machines for Regression*, ECML

[3] Lei et al. (2018), *Distribution-Free Predictive Inference for Regression*, JRSS B

Split Conformal Prediction (SCP)[1,2,3]: toy example — prediction step



- Predict with $\hat{\mu}$
- Build $\widehat{C}_\alpha(x)$: $[\hat{\mu}(x) \pm q_{1-\alpha}(\mathcal{S})]$

Back to SCP

[1] Vovk et al. (2005), *Algorithmic Learning in a Random World*

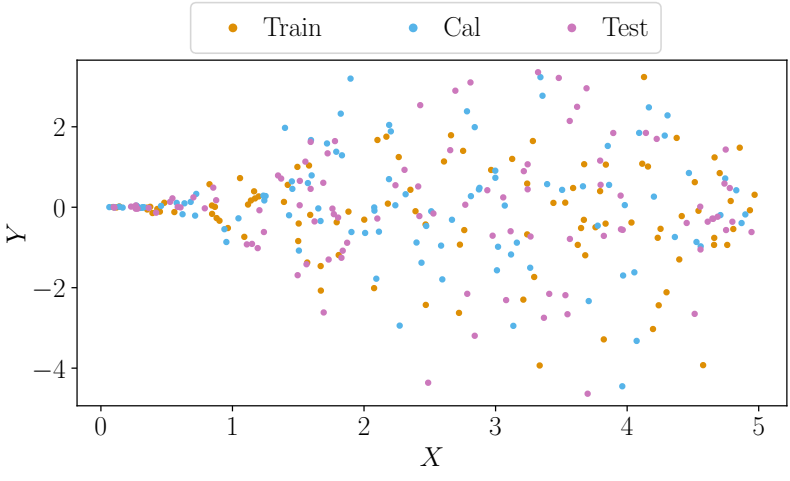
[2] Papadopoulos et al. (2002), *Inductive Confidence Machines for Regression*, ECML

[3] Lei et al. (2018), *Distribution-Free Predictive Inference for Regression*, JRSS B

SCP

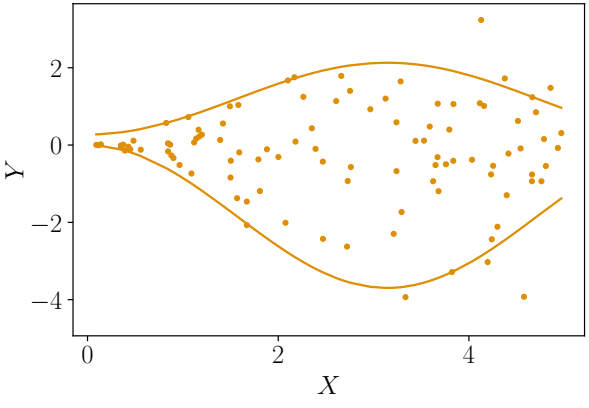
CQR

Conformalized Quantile Regression (CQR)[5]



[5] Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

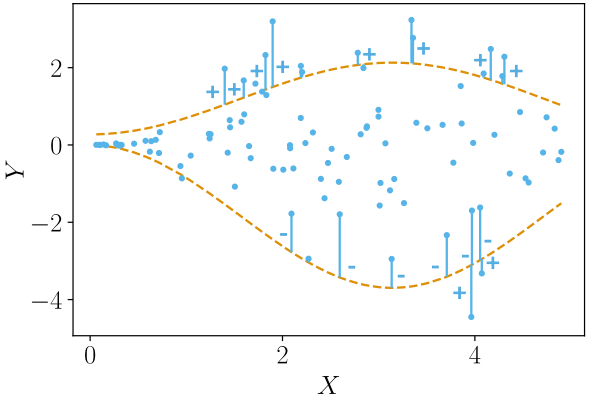
Conformalized Quantile Regression (CQR)[5] — training step



- Learn (or get) $\widehat{\mathrm{QR}}_{\mathrm{lower}}$ and $\widehat{\mathrm{QR}}_{\mathrm{upper}}$

[5] Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS

Conformalized Quantile Regression (CQR)[5] — calibration step



- Predict with $\widehat{\mathrm{QR}}_{\mathrm{lower}}$ and $\widehat{\mathrm{QR}}_{\mathrm{upper}}$
- Get the scores $\mathcal{S} = \{S_i\}_{\mathrm{Cal}} \cup \{+\infty\}$
- Compute the $(1-\alpha)$ empirical quantile of $\mathcal{S}$, noted $q_{1-\alpha}(\mathcal{S})$

$$\hookrightarrow \quad S_i := \max\left\{\widehat{\mathrm{QR}}_{\mathrm{lower}}(X_i) - Y_i,\, Y_i - \widehat{\mathrm{QR}}_{\mathrm{upper}}(X_i)\right\}$$

Back to CQR

[5] Romano et al. (2019), *Conformalized Quantile Regression*, NeurIPS